TT-P-605B
July 29, 1976
SUPERSEDING
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April 27, 1966

FEDERAL SPECIFICATION

PRIMER COATING, ALL PURPOSE GRAY, ALKYD (IN PRESSURIZED DISPENSERS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers an all purpose light gray primer in a pressurized dispenser for interior and exterior application for use on metal, wood, and plastic.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-C-282 - Chipboard.
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8010

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 609 - Methods for Preparation of Steel Panels for Testing Paint, Varnish, Lacquer and Related Products.
D 2369 - Method of Test for Volatile Content of Paints.
D 3335 - Determination of Low Concentrations of Lead in Paint by Atomic Absorption Spectroscopy.
D 1308 - Test for Effect of Household Chemicals on Clean and Pigmented Organic Finishes.

Chemical Specialties Manufacturers Association (CSMA) Publications:

CSMA - Aerosol Guide.

Revised Flammability Test Methods for Aerosol Products (Adopted September 12, 1963), Sixth Edition, March 1971.

(Application for copies should be addressed to Chemical Specialties Manufacturers Association, Inc., 1001 Connecticut Ave, N.W., Washington, DC 20036.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Composition.

3.1.1 The primer shall be composed predominantly of an alkyd resin, suitable white pigments, lampblack, solvents, and propellants to meet all requirements of this specification.

3.1.2 The propellant shall be a hydrocarbon, a halogenated hydrocarbon, or a blend of hydrocarbons and/or halogenated hydrocarbon(s) as required to assure that the primer conforms to the requirements as specified herein. Vinyl chloride shall not be used in these products.

3.2 Dispenser.

3.2.1 Container. The container shall be commercial-type metal pressure container generally known as an aerosol container of such construction as to assure acceptance of the finished package by common carriers operating in interstate commerce. The capacity of the container shall be sufficient to contain the specified net contents of the product with outage in accordance with generally recognized safe commercial practice. Each container shall be labeled as stated in 5.3.3.

3.2.2 Dispensing valve. The valve shall have a spray head which can be removed without releasing pressure from the aerosol. The valve shall have a metering orifice accessible to cleaning with a common pin after removal of the spray head. The removable spray head shall contain orifices of such dimensions as to produce spraying properties as required in 3.6.2. The lower end of the dip tube shall have a maximum clearance of 9.5 mm (3/8 inch) from the bottom rim of the can after storage for one week in the product (which allows for swelling of the tube).

3.2.3 Agitator. Each dispenser of the pigmented material shall contain one or more agitators capable of dispersing the pigment by normal shaking. The agitators shall be made of materials which shall not react with or contaminate the primer or container.

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3.2.4 Cover cap. The valve shall be protected from accidental functioning and damage by a rigid cover cap which shall be removable and replaceable. The color of the cover cap shall be an approximate match to the color of the primer in the container.

3.3 Quantitative requirements.

3.3.1 Net contents. Unless otherwise specified, depending upon the propellant used, each can shall contain a minimum weight of 13 avoir. ounces and a maximum of 16 avoir. ounces at the discretion of the supplier, and the unit shall be a nominal 1 pint can. Net contents shall be determined as specified in 4.4.2.

3.3.2 Total solids. Regardless of net weight, the container shall have a minimum amount of solids per can which shall be not less than 78 g (2.75 oz)/can, when calculated as in 4.4.3.

3.3.3 Lead content. The primer shall contain not more than one-half of one percent lead by weight of nonvolatile when tested as specified in 4.4.4.

3.3.4 Pressure. When tested as specified in 4.4.5 the aerosol container shall not burst or show evidence of leakage, distortion, or any other defects at an equilibrium pressure which shall not exceed 140 p.s.i.g. at 55 ± 1°C (130 ± 2°F). The samples for the purpose of this test shall be selected in accordance with level S-3 of MIL-STD-105. Any failure of this test shall be cause for rejection of the batch.

3.4 Storage stability. At all times during the year following the date of acceptance, the aerosols must be useable. Sampling shall be in accordance with 4.2, Inspection Level with 4.3 and AQL level in accordance with 4.3.1. Useability is determined by:

(a) Spraying properties (see 3.5.2).
(b) Condition in container (see 3.5.1).
(c) Color (see 1.1 and 6.3).
(d) Drying time (see 3.5.6).
(e) Pressure (see 3.3.4).
(f) Valve operation (see 3.5.4).

3.5 Qualitative requirements.

3.5.1 Condition in container. The primer shall be easily redispersed by normal shaking. There shall be no seeding or other agglomeration when tested as in 4.4.6.

3.5.2 Spraying properties. When tested as specified in paragraph 4.4.7, the content of the aerosol container shall be sprayable when stabilized at temperatures from 15 to 32°C. (60 to 90°F). A minimum of 98 percent by weight of content shall be sprayable in a pattern such as to deposit a uniform coating on a smooth vertical surface. The film shall dry free of orange peel, sags, specks, or any other surface irregularities. Leftover lacquer of more than 2 percent shall constitute a defect.

3.5.3 Immersion resistance. Immediately upon removal of the panel from distilled water the film shall show no blistering or other defects. There shall be no softening of the film after 24 hours' drying, when tested as in 4.4.8.

3.5.4 Valve operation. The valve shall operate freely without excessive finger pressure. It shall close immediately upon release of finger pressure. The spray shall be continuous without sputtering or interruption. The valve shall not clog nor collect heavy deposits of coating around the orifice when tested as in 4.4.9.

3.5.5 Flexibility (cold cracking). The primer film shall show no cracking or loosening of film from the metal when prepared and tested as described in 4.4.10.

3.5.6 Drying time. A sprayed coat of primer shall dry dust free within 5 minutes, shall be tack free within 10 minutes, and shall dry through within 2 hours, when tested as specified in 4.4.11.

3.5.7 Self-lifting. There shall be no self-lifting when recoated at 1 and 4 hours, when tested as specified in 4.4.12.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. Sampling for visual examination and testing shall be in accordance with MIL-STD-105.

4.2.1 Visual examination. The inspection level shall be S-4, with an acceptable quality level (AQL) of 2.5 percent defective. Check for the following: Leaks, rust, damage (dents, bruises, scratches), film defects, improper marking, ease of operation (shake and spray), and other deficiencies which may impair serviceability.

4.2.2 Tests. Test shall be performed as indicated in 4.4. Except as otherwise specified, inspection level S-3, AQL 2.5 percent defective, shall be used for tests performed on end items. Inspection level S-3 shall be used for the pressure test (3.3.1), and failure of any one sample to meet this test shall be cause for rejection of the lot.

4.3 Inspection. Inspection shall be performed in accordance with method 1011 of Fed. Test Method Std. No. 141.

4.3.1 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5. Defectives shall be scored in accordance with table I. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 percent defective.

4.3.2 Examination of defects. Examination of defects shall be as outlined in table I.

TABLE I. Classification of preparation for delivery defects

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior and interior container)	Number per container is more or less than required. Net weight exceeds requirements.

4.4 Test procedures. Alternate test techniques and equipment may be used, but in case of dispute, the test methods specified here shall prevail. Unless otherwise specified, standard test conditions are 23 ± 1°C (73 ± 2°F) and a relative humidity of 50 ± 5 percent.

4.4.1 Test concentrate. With the container well agitated and in an upright position, pierce the top with a needle point, making a very fine hole which will allow the propellant to escape (conditioning at low temperature for 1 hour will facilitate this). After a few minutes, the pierced hole can be enlarged, and as soon as there is no further evidence of gas escaping, the top of the container can be cut open. Allow to stand until the contents are at room temperature, then transfer the contents quantitatively into a tared beaker using several washes of acetone. Evaporate on a water bath until no more propellant is liberated and until the contents are less than 40 percent of the original weight of the contents including propellant. Then read just to exactly the following percent of original weight with n-butyl acetate:

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Nominal can contents	Percent
avoir. ounces	
16	40.0
15	42.6
14	45.7
13	49.2

This compensates for the differences in solids for cans of different nominal net contents and makes all test concentrates equivalent for further testing.

4.4.1.1 For the qualitative tests, the manufacturer may use the primer concentrate as it exists before packaging and pressurization, provided it is adjusted as specified in 4.4.1.

4.4.2 Total net contents. The total net contents shall be determined by calculating the difference in weight between the filled container and the completely empty one (see 4.4.2 for procedure of emptying). Wash out interior of can with acetone and dry. (Note that the weight of the agitators must be included in the tare.) Using the completely emptied container as a tare, additional checks can be made on filled containers. At least two containers shall be checked from each lot sampled since variations in weight of empty cans can be found. A check on full and empty cans shall be made on the same cans and conformance with 3.3.1 determined.

4.4.3 Total solids. Using the test concentrate (4.4.1) adjusted as noted, determine the nonvolatile content in accordance with ASTM method D 2369. The total solids shall then be calculated by multiplying the nonvolatile figure by the percent to which the test concentrate was adjusted and dividing by 100. Check for compliance with 3.3.2.

4.4.4 Lead content. The lead content shall be determined in accordance with ASTM D 3335, and conformance with 3.3.3 determined.

4.4.5 Pressure. Containers filled for shipment shall be submerged in a water bath maintained at 55 ± 1°C (130 ± 2°F). The pressure shall be measured after 15 minutes of immersion in accordance with the method described in the CSMA Aerosol Guide, "Methods for Pressure Measurement in Metal Aerosol Containers", method Z. Evaluate for conformance with the requirements specified in 3.3.4.

4.4.6 Condition in container. Observe primer for conformance to 3.5.1 while testing spraying properties (4.4.7).

4.4.7 Spraying properties. Spray a 10- by 30-cm (4- by 12-inch) steel panel prepared in accordance with ASTM method 609, (Type I, method 5B). Set the panel at an angle of approximately 15° from the vertical. Hold the dispenser in an upright parallel position to the panel at a distance of 25 to 30 cm (10 to 12 inches) and with the valve fully open, move the dispenser horizontally from one side of the panel to the other using a stroking motion. Spray in overlapping horizontal strokes from the top to the bottom of the panel. Apply two thin coats (each coat approximately 0.012 mm (0.005 inch) dried film thickness) one after the other. The second thin coat can be applied by spraying vertically in overlapping strokes from side to side. After application of the second thin coat, place the panel in a vertical position, to dry for 5 minutes, and examine the film for smoothness of flowout, freedom from specks, bubbles, craters or any apparent abnormalities.

Continue spraying the content to completion on a flat surface and note the spraying properties at intervals. Calculate the leftover lacquer as follows:

$$\text{Percent of leftover lacquer} = \frac{C}{A-B} \times 100$$

Where: A = weight in grams of full can.
B = weight in grams of empty can.
C = weight in grams of leftover lacquer.

Check the results for compliance with the requirements of paragraph 3.5.2.

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4.4.8 Immersion resistance. Using a film applicator having a 0.015-mm (0.006-inch) gap clearance, draw down, in duplicate, a film of the test concentrate (4.4.1) on a panel specified in ASTM D 609 (Type I, method 5B). Air dry for 18 hours, then immerse in distilled water at room temperature for 24 hours. Check for compliance with 3.5.3.

4.4.9 Valve operation. Spray from the dispenser with the valve fully open for 15 seconds each time, once per day, on any 7 days over a period of 10 days. Observe performance for compliance with 3.5.4. (After each spraying period, the valve shall be cleared according to manufacturer's directions.)

4.4.10 Flexibility (cold cracking). The flexibility tests shall be as determined by method 6223 of Fed. Test Method Std. No. 141. The panel shall be as specified in method 2012, paragraph 1.2. The method of application shall be by drawing down the test concentrate (4.4.1) with a film applicator and the dried film thickness shall be 0.025 $\pm$ 0.005 mm (0.001 $\pm$ 0.0002 inch). The panel shall be air-dired at room temperature for 48 hours. Bend over a 6.25-mm (1/4-inch) diameter rod and determine compliance with 3.5.5.

4.4.11 Drying time. Prepare a panel as specified in method 2012, paragraph 1.2, of Fed. Test Method Std. No. 141 and spray with primer to a dry film thickness of approximately .025mm (0.001 inch). Test for drying time according to method 4061 of Fed. Test Method Std. No. 141 and examine for compliance with 3.5.6.

4.4.12 Self-lifting. The primer shall be sprayed on a metal panel (ASTM D 609, Type I, Method 5B) to a dried film thickness of approximately 0.012 mm (0.0005 inch). The panel shall be recoated at 1 hour and 4 hours and tested according to method 6252 of Fed. Test Method Std. 141. Determine compliance with 3.5.7.

5. PREPARATION FOR DELIVERY

5.1 Packaging. (See 3.2.)

5.2 Packing.

5.2.1 Level A. Twelve dispensers of the same description shall be packed in a close-fitting box conforming PPP-B-636, V3c; or PPP-B-640, class 2. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. The gross weight of the triplewall fiberboard box shall not exceed 200 pounds. The gross weight of the V3c box shall not exceed the weight limitations of PPP-B-636.

5.2.2 Level B. Twelve dispensers of the same description shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. Separators the full height of the box shall be required conforming to UU-C-282, 0.040 in. minimum thickness. The box shall be closed in accordance with the appendix to PPP-B-636.

5.2.3 Level C. The primer, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to destination. Containers shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil agencies. Packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

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5.3.3 Special marking. Legible commercial marking shall be acceptable. Cans overcoated to obliterate original marking shall not be acceptable. Marking shall be lithographed on the unit container or printed on a label securely affixed to the container and shall include the following:

(a) Color number of Fed. Std. No. 595.
(b) Net weight in ounces.
(c) Name of material.
(d) Directions for use including surface preparation.
(e) A formula in general terms.
(f) For best results use at temperatures between 16 and 32°C (60° to 90°F).
(g) Date of manufacture.
(h) Do not puncture or incinerate.
(i) Keep from fire or flame.
(j) Use with adequate ventilation.
(k) Avoid repeated contact with skin.
(l) Store below 49°C (120°F).
(m) Keep from children.

6. NOTES

6.1 Intended use. This primer in pressurized dispensers provides a quick convenient method for priming small objects or areas with a general purpose coating. It can be applied on metal, wood, and plastic. Surfaces to be sprayed should be clean, dry, and free from oily or greasy films.

6.2 Net contents. The unit can will be known as a 1-pint can. Unless otherwise specified, it shall be understood that the weight of the contents may be from 13 to 16 avoir. ounces, at the discretion of the supplier, provided the total solids of the contents supplied is in accordance with 3.3.2. The purpose of this is to permit wide latitude in formulation while assuring uniform value of each unit based on paint solids. The net weight therefore need not be specified.

6.3 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Availability of color chips if specified.
(d) Size of container (pint) unless otherwise specified.
(e) Quantities of unit packages in shipping container (see 5.2.1).

Civil Agency Coordinating Activities:

DOT-FAA
NASA-JFK

Preparing activity:

GSA-FSS

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